

**Series AAB5/5****SET No. 3**रोल नं.  
Roll No. 

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प्रश्न पत्र कोड  
Q.P. Code**56/5/3**

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 11 हैं।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को छात्र उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में 12 प्रश्न हैं।
- कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
- Please check that this question paper contains 11 printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains 12 questions.
- Please write down the Serial Number of the question in the answer-book before attempting it.
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

## रसायन विज्ञान (सैद्धान्तिक) CHEMISTRY (Theory)

निर्धारित समय : 2 घण्टे  
Time allowed : 2 hoursअधिकतम अंक : 35  
Maximum Marks : 35

56/5/3

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**General Instructions :**

Read the following instructions very carefully and strictly follow them:

1. This question paper contains **12** questions. **All** questions are compulsory.
2. This question paper comprises of **three** sections- **Section A, B and C**.
3. **Section A** – Q. No. 1 to 3 are very short-answer type questions carrying **2** marks each.
4. **Section B** – Q. No. 4 to 11 are short-answer type questions carrying **3** marks each.
5. **Section C** – Q. No. 12 is case based question carrying **5** marks.
6. Use of log tables and calculators is **not** allowed.

**SECTION A**

1. Explain the following reactions : 1×2=2
  - (a) Wolff Kishner reduction
  - (b) Cannizzaro reaction
2. Arrange the following in the increasing order of the property mentioned : 1×2=2
  - (a)  $\text{CH}_3\text{COOH}$ ,  $\text{ClCH}_2\text{COOH}$ ,  $\text{FCH}_2\text{COOH}$  (Acid strength)
  - (b)  $\text{CH}_3\text{CHO}$ ,  $\text{CH}_3\text{CH}_2\text{OH}$ ,  $\text{CH}_3\text{CH}_2\text{CH}_3$  (Boiling Points)
3. Answer the following questions (Do any two) : 1×2=2
  - (a) Identify the order of reaction from the following unit for its rate constant :  
 $\text{Lmol}^{-1}\text{s}^{-1}$
  - (b) The conversion of molecules A to B follow second order kinetics. If concentration of A is increased to three times, how will it affect the rate of formation of B?
  - (c) Write the expression of integrated rate equation for zero order reaction.



## SECTION B

4. (a) Account for the following : 1×3=3

- (i)  $pK_b$  of aniline is more than that of methylamine.
- (ii) Aniline does not undergo Friedel-Crafts reaction.
- (iii) Primary amines have higher boiling points than tertiary amines.

**OR**

- (b) (i) Arrange the following compounds in the increasing order of their basic strength in aqueous solution : 1×3=3  
 $CH_3NH_2$ ,  $(CH_3)_3N$ ,  $(CH_3)_2NH$
- (ii) What is Hinsberg's reagent ?
  - (iii) What is the role of pyridine in the acylation reaction of amines?

5. (a) Write the IUPAC name of the following complex : 1×3=3  
 $K_3[Cr(C_2O_4)_3]$

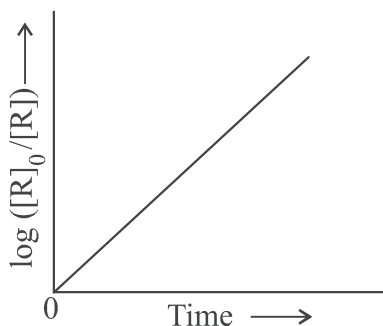
- (b) On the basis of crystal field theory, write the electronic configuration of  $d^5$  ion if  $\Delta_0 < P$ .

- (c) What are ambidentate ligands ?

6. (a) Write any two consequences of Lanthanoid Contraction. 2+1=3

- (b) Name the element of 3d series which exhibits the largest number of oxidation states. Give reason.

7. Observe the graph shown in figure and answer the following questions : 1×3=3



- (a) What is the order of the reaction ?
- (b) What is the slope of the curve?
- (c) Write the relationship between  $k$  and  $t_{1/2}$  (half life period).

8. Give reasons for the following statements : 1×3=3
- (a) Transition elements and their compounds act as good catalysts.
- (b)  $E^\circ (\text{Mn}^{2+}/\text{Mn})$  value is highly negative as compared to other elements.
- (c)  $\text{Cr}^{2+}$  is a strong reducing agent.

9. (a) (i) Define coagulation. 1×3=3
- (ii) State Hardy-Schulze rule.
- (iii) What is Electrophoresis?

**OR**

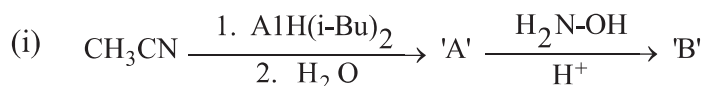
- (b) Write three differences between Physisorption and Chemisorption. 1×3=3

10. (a) An aromatic compound 'A' ( $\text{C}_7\text{H}_6\text{O}_2$ ) on reaction with aqueous ammonia and heating forms compound 'B'. 'B' on heating with  $\text{Br}_2$  and alcoholic potash forms a compound 'C' of molecular formula  $\text{C}_6\text{H}_7\text{N}$ . Write the reactions involved and identify 'A', 'B', 'C'. 3

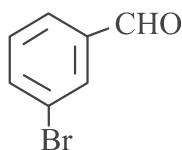
**OR**

- (b) A compound 'A' on reduction with iron scrap and hydrochloric acid gives compound 'B' with molecular formula  $\text{C}_6\text{H}_7\text{N}$ . Compound 'B' on reaction with  $\text{CHCl}_3$  and alcoholic KOH produces an obnoxious smell of carbylamine due to the formation of 'C'. Identify 'A', 'B' and 'C' and write the chemical reactions involved. 3

11. (a) Complete the following : 1×3=3



- (ii) Write IUPAC name of the following compound:



- (iii) Write chemical test to distinguish between the following compounds :  
Phenol and Benzoic acid

**OR**

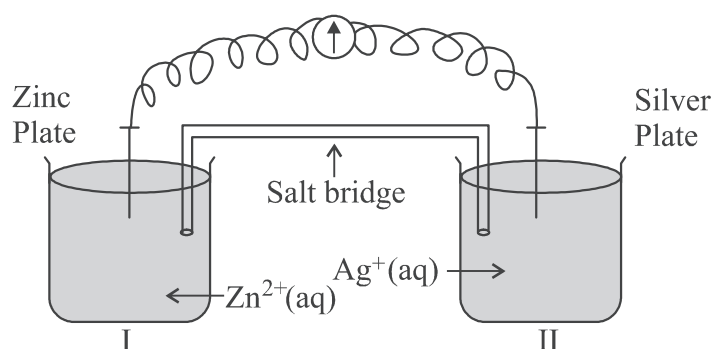
- (b) Convert the following : 1×3=3
- (i) Benzoic acid to Benzaldehyde
- (ii) Propan-1-ol to 2-Bromopropanoic acid
- (iii) Acetaldehyde to But-2-enal

## SECTION C

12. Read the passage given below and answer the questions that follow : 1+1+1+2=5

Oxidation-reduction reactions are commonly known as redox reactions. They involve transfer of electrons from one species to another. In a spontaneous reaction, energy is released which can be used to do useful work. The reaction is split into two half reactions. Two different containers are used and a wire is used to drive the electrons from one side to the other and a Voltaic/Galvanic cell is created. It is an electrochemical cell that uses spontaneous redox reactions to generate electricity. A salt bridge also connects to the half cells. The reading of the voltmeter gives the cell voltage or cell potential or electromotive force. If  $E_{\text{cell}}^{\circ}$  is positive the reaction is spontaneous and if it is negative the reaction is non-spontaneous and is referred to as electrolytic cell. Electrolysis refers to the decomposition of a substance by an electric current. One mole of electric charge when passed through a cell will discharge half a mole of a divalent metal ion such as  $\text{Cu}^{2+}$ . This was first formulated by Faraday in the form of laws of electrolysis.

The conductance of material is the property of materials due to which a material allows the flow of ions through itself and thus conducts electricity. Conductivity is represented by  $k$  and it depends upon nature and concentration of electrolyte, temperature etc. A more common term molar conductivity of a solution at a given concentration is conductance of the volume of solution containing one mole of electrolyte kept between two electrodes with the unit area of cross-section and distance of unit length. Limiting molar conductivity of weak electrolytes cannot be obtained graphically.



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- (a) Is silver plate the anode or cathode? 1
- (b) What will happen if the salt bridge is removed? 1
- (c) When does electrochemical cell behaves like an electrolytic cell? 1
- (d) (i) What will happen to the concentration of  $\text{Zn}^{2+}$  and  $\text{Ag}^+$  when  $E_{\text{cell}} = 0$ . 1×2=2
- (ii) Why does conductivity of a solution decreases with dilution?

**OR**

- (d) The molar conductivity of a 1.5 M solution of an electrolyte is found to be 138.9  $\text{S cm}^2 \text{mol}^{-1}$ . Calculate the conductivity of this solution. 2

